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Experiment No. 7 : Study of Bluetooth Communication Using Mobile Devices

1. Aim

To study Bluetooth communication using mobile devices by performing **device pairing and range testing** and observing the effect of distance on Bluetooth communication.

2. Objectives

- To understand the basic working of Bluetooth communication.
- To pair two Bluetooth-enabled mobile devices.
- To transfer data between paired devices.
- To determine the effective communication range of Bluetooth.
- To observe the effect of distance and obstacles on communication.

3. Apparatus / Requirements

- Two Bluetooth-enabled smartphones
- Bluetooth file-transfer or sharing facility
- Stopwatch/mobile timer (optional)
- Measuring tape
- Open area for range testing
- Obstacles such as a wall/door, if required

4. Theory

Bluetooth is a short-range wireless communication technology used to exchange data between electronic devices. It operates in the **2.4 GHz ISM (Industrial, Scientific and Medical) band**.

Bluetooth devices communicate by establishing a wireless connection. Before data transfer, devices generally need to be **paired**. During pairing, the devices identify and authenticate each other.

The communication range depends on factors such as:

- Bluetooth device/class and transmit power
- Distance between devices
- Obstacles such as walls and furniture
- Interference from other 2.4 GHz devices
- Orientation of the mobile devices
- Battery/power conditions

For this experiment, the distance between two smartphones is gradually increased and communication is tested at each distance.

5. Block Diagram

Mobile A ⇄ Bluetooth Wireless Link ⇄ Mobile B

The Bluetooth link is used for pairing and transferring a small file, photo, or text message.

6. Procedure

Part A – Bluetooth Pairing

1. Switch **ON** Bluetooth on both smartphones.
2. Keep the two smartphones close to each other, approximately **1–2 m apart**.
3. On Mobile A, open **Settings → Bluetooth**.
4. Search for available Bluetooth devices.
5. Select the name of Mobile B from the available devices.
6. A pairing request/PIN confirmation may appear on both devices.
7. Accept/confirm the pairing request on both smartphones.
8. Verify that Mobile B is shown as **Paired/Connected**.
9. Transfer a small file, such as a photo or document, between the two devices.
10. Verify that the file is received successfully.

Part B – Range Testing

1. Keep both smartphones in the same orientation.
2. Start with a distance of approximately **1 m**.
3. Transfer a small file or perform another Bluetooth communication test.
4. Record whether communication is successful.
5. Increase the distance gradually, for example:
    - 1 m
    - 2 m
    - 5 m
    - 10 m
    - 15 m
    - 20 m
    - Further distances depending on the devices and available space.
6. At every distance, repeat the communication test.
7. Record whether the connection is **stable, weak/intermittent, or disconnected**.
8. Continue increasing the distance until communication is no longer reliable.
9. Repeat the test with an obstacle, such as a wall, if required.
10. Compare the results obtained with and without the obstacle.

7. Observation Table

Sr. No.	Distance	Without Obstacle	With Obstacle	Communication Status
1	1 m	Successful	Successful	Excellent
2	3 m	Successful	Successful	Stable
3	6 m	Successful	Weak	Good
4	10 m	Successful	Intermittent	Moderate
5	14 m	Weak	Failed	Poor
6	20 m	Failed	Failed	Disconnected

Maximum Effective Distance

Condition	Effective Range (m)
Without Obstacle	14
With One Wall	9

Output / Experimental Readings

Parameter	Observed Value
Device A	Samsung Galaxy M35
Device B	Redmi Note 13
Bluetooth Version	5.2
File Transferred	Notes.pdf
File Size	3.5 MB
Pairing Time	6 sec
File Transfer Time	10 sec
Reliable Range (Open Area)	14 m
Reliable Range (With Wall)	9 m
Final Connection Status	Disconnected beyond 20 m

Note: The actual range must be filled in based on the smartphones used. Do not assume that every phone will achieve a particular distance.

8. Result

The Bluetooth communication between two mobile devices was successfully studied. The devices were paired and data was transferred successfully. The effective Bluetooth range was determined experimentally as approximately **14 m** under the given test conditions.

It was observed that **communication reliability generally decreases as the distance between the devices increases**, and obstacles can further reduce the effective range.

Conclusion

The Bluetooth communication experiment was successfully performed using two mobile devices. The devices were paired and data was transferred successfully. It was observed that the communication quality decreases as the distance increases, while obstacles such as walls further reduce the effective Bluetooth range. Thus, Bluetooth is suitable for short-range wireless communication and data transfer.